

Lecture 11

EF Core Relationships and Migrations

Continuation of the layered Expense Tracker

Every foreign key has a business meaning before it has a database type.

- Model User - Expense and Category - Expense relationships.
- Use foreign keys, navigation properties, Include and migrations.
- Replace disposable schema creation with controlled history.

Domain: entities plus navigation properties.

Application: use cases.

Infrastructure: Fluent API mapping and migrations.

Presentation: no connection strings.

Project: migration startup host.

One user has many expenses.

One category has many expenses.

Migrations describe schema changes.

Never rewrite an applied shared migration.

- Read the runnable CodeExamples project first.
- Run the Project and identify where each responsibility lives.
- Change one implementation detail without breaking the Application layer.
- Write a short explanation of the boundary you protected.

- Apply the same decision to the student's chosen domain.
- Keep a screenshot or short note for the evolution portfolio.
- Use the database-session model and constraints.
- Prepare one demonstration scenario for the final defense.